Master of Computer Applications (MCA) - Final Year Syllabus
Course Code: MCA-501: Advanced Cloud Computing and Distributed Systems
Academic Year: 2026-2027

Unit 1: Distributed Architecture Fundamentals
Distributed computing enables disparate machines to coordinate over a computer network.
The primary objectives of distributed systems are scalability, fault tolerance, and transparency.
Consensus algorithms like Raft and Paxos ensure synchronized state machine replication across distributed nod
In Raft, leader election is triggered via randomized election timeouts ranging between 150ms and 300ms.
Distributed hash tables (DHT) like Chord utilize consistent hashing with SHA-1 to map keys to nodes efficiently.

Unit 2: Microservices and Storage Systems

Microservices architecture decomposes monolithic applications into independent, loosely coupled services.

Communication between microservices typically uses gRPC for high performance internal RPC or RESTful HTT

CAP theorem proves that a distributed database can only provide at most two out of three guarantees: Consistency, Availability, and Partition Tolerance.

NoSQL databases such as Apache Cassandra prioritize Availability and Partition Tolerance (AP systems), utilizing eventual consistency with tunable read and write consistency levels.

Unit 3: Final Examination Criteria and Project Grading

Final project assessment comprises 100 total marks broken into three components:

1. System Architecture and Implementation Quality: 50 Marks
2. Automated Test Suite and Code Documentation: 25 Marks
3. Viva Voce and Live Demonstration: 25 Marks

Minimum passing grade is 50% aggregate score. Plagiarism exceeding 15% results in immediate disqualificatio

Final project reports must be submitted in IEEE conference format before December 15th.